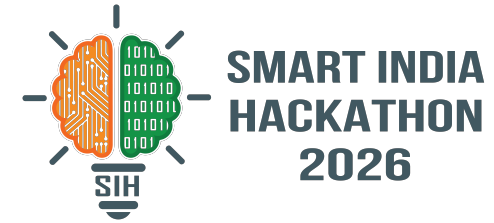
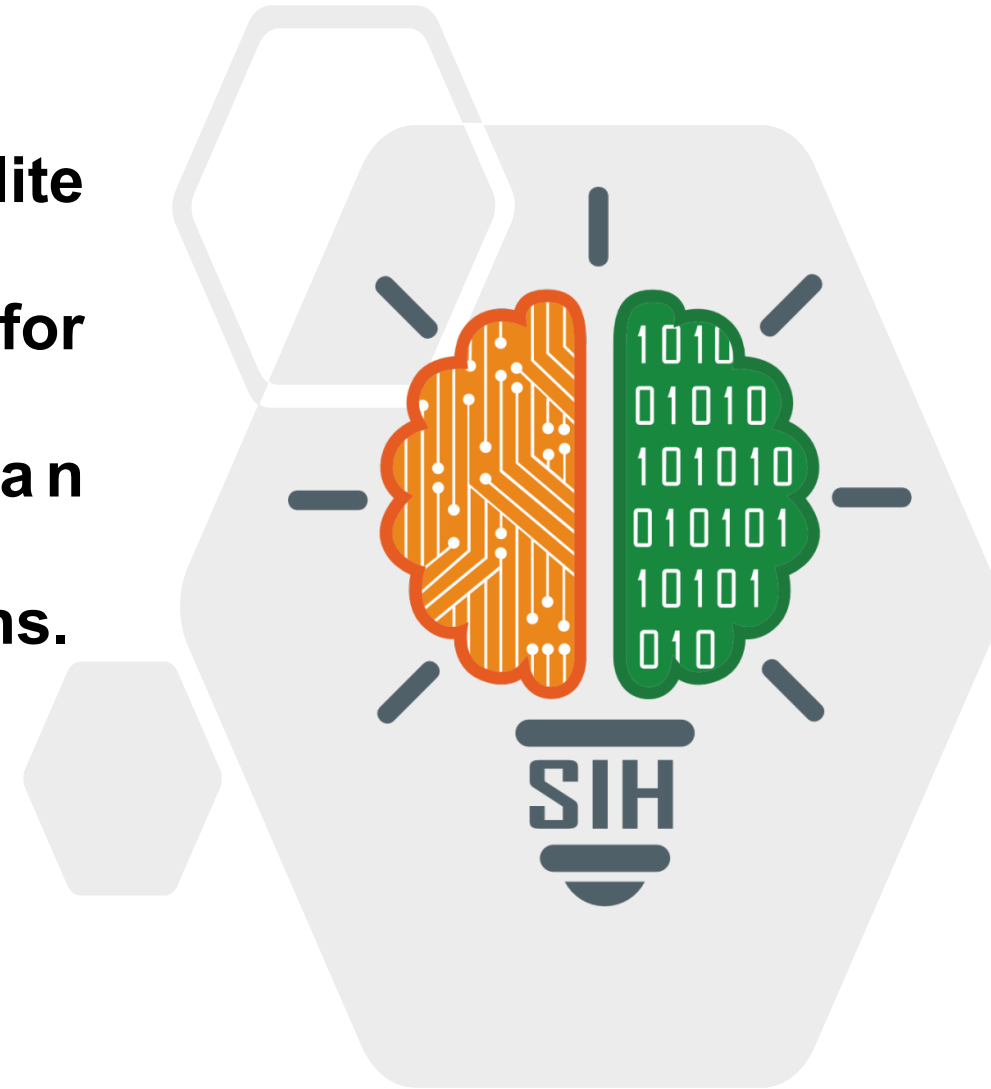


SMART INDIA HACKATHON 2026



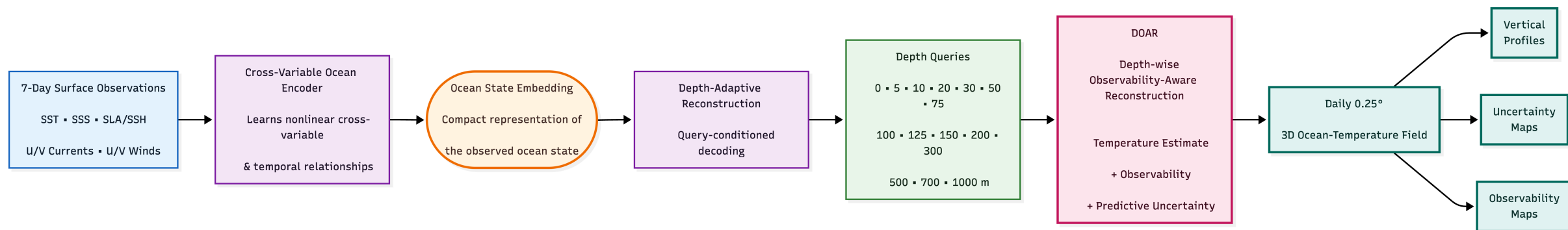
- **Problem Statement ID – SIH26066**
- **Problem Statement Title – OceanEmbed: Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.**
- **Theme – Disaster Management**
- **PS Category – Software**
- **Team Name – Neutrons.**



Problem

Satellite observations continuously describe the **ocean surface**, while subsurface temperature remains sparsely observed by Argo floats and other in-situ systems. SIH26066 requires daily 0.25° reconstruction of the North Indian Ocean at 15 standard depths.

Our approach

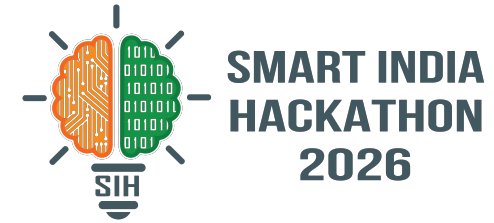


Innovation & Uniqueness

Instead of asking only “What is the temperature at this depth?”, OceanEmbed-X also asks “**How strongly can the available surface observations constrain this depth?**”

TECHNICAL APPROACH

Neutrons



① Surface State

$$X_{t-6:t} = [\text{SST}, \text{SSS}, \text{SLA}, U, V, U_w, V_w]$$

7-day multi-source surface observations

② Ocean Embedding → Depth Reconstruction

$$Z = f_{\theta}(X_{t-6:t})$$

$$\hat{T}(z) = g_{\phi}(Z, q_z, E_z)$$

Latent ocean state + depth-specific query + vertical prior

③ DOAR + Physics

$$\{\hat{T}(z), O(z), \sigma(z)\}$$

$$L = L_{\text{data}} + \lambda_1 L_{\text{surface}} + \lambda_2 L_{\text{vertical}} + \lambda_3 L_{\text{temporal}} + \lambda_4 L_{\text{range}}$$

Prediction + observability + uncertainty

1. DATA HARMONISATION

SST • SSS • SLA/SSH
U/V Current • U/V Wind
0.25° • Daily

2. OCEAN STATE ENCODER

Variable Encoding
Cross-Attention
7-Day Temporal Memory

OCEAN STATE EMBEDDING

Compact Latent
Ocean Representation

3. DEPTH-ADAPTIVE DECODER

EOF / Vertical Prior
15 Depth Queries
0 → 1000 m

4. DOAR

$\hat{T}(z)$ • $O(z)$ • $\sigma(z)$
Temperature •
Observability • Uncertainty

5. PHYSICS-AWARE TRAINING

Surface Anchor
Vertical Regularity
Temporal Consistency
Physical Range

OUTPUT

DAILY 3D TEMPERATURE
+ OBSERVABILITY
+ UNCERTAINTY

Feasibility

The required workflow can be built using established open scientific-data tooling:

Python + PyTorch + Xarray + Dask + NetCDF

Training / reconstruction target:
Copernicus GLORYS12V1

Surface observations:
Copernicus Marine satellite/reprocessed products

Validation:
Held-out **Argo profiles**, using leakage-controlled temporal/spatial splits.

SIH itself identifies GLORYS and INCOIS/Argo resources as recommended data sources and permits regridding/interpolation when required

Risks → Solutions

Multi-source grid/time mismatch
→ Xarray regridding + UTC daily harmonisation

Missing observations
→ quality masks + mask-aware fusion

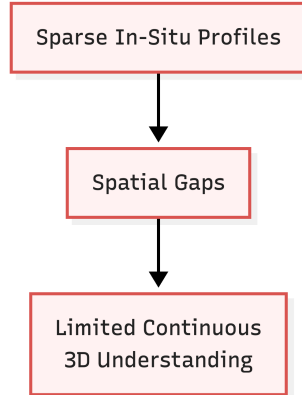
Weak deep-ocean predictability
→ DOAR + vertical prior + explicit uncertainty

Large datasets / GPU limits
→ Dask chunking + patch-based training + mixed precision

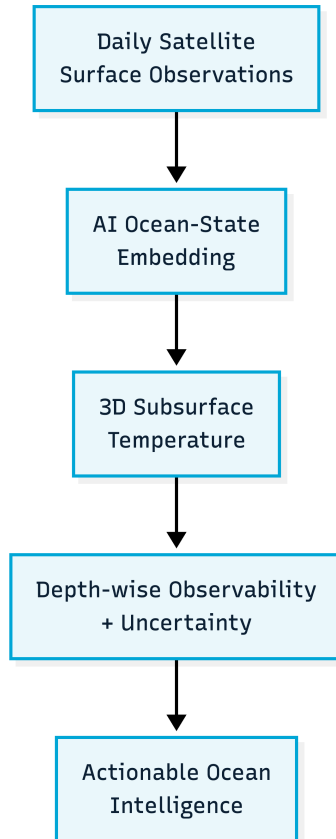


FROM SPARSE PROFILES TO A CONTINUOUS, CONFIDENCE-AWARE OCEAN INTERIOR

TODAY



WITH OCEANEMBED-X



WHO BENEFITS?

MoES / INCOIS

Large-scale continuous subsurface ocean-state information.

Ocean & Climate Researchers

Improved spatial/temporal reconstruction for studying circulation, heat content and air-sea interaction.

Marine Heatwave / Cyclone Monitoring

Subsurface thermal information supporting ocean-heat-content and thermal-structure analysis.

Fisheries & Ecosystem Applications

Improved understanding of changing thermal conditions.

Observation planning

Low-observability regions can be highlighted as potential **high-value locations for additional in-situ observations.**

“OceanEmbed-X does not replace Argo — it identifies where satellite observations can fill the gaps, and where direct measurements remain essential.”

Existing work

- Deep learning can reconstruct subsurface ocean temperature from satellite surface observations.
- Recent methods incorporate **temporal information, physical/vertical priors and uncertainty estimation**.
- However, **predictability decreases where surface observations contain weak information about deeper layers**.

RESEARCH GAP

Surface observations do not constrain every depth equally.

OUR CONTRIBUTION — DOAR

Depth-wise Observability-Aware Reconstruction

For every **location × depth**, OceanEmbed-X estimates:

- $\hat{T}(z)$ — Temperature
- $O(z)$ — Observability Index
- $\sigma(z)$ — Predictive Uncertainty

CORE REFERENCES & DATA

- [1] SIH 2026 — **SIH26066, MoES / INCOIS**
- [2] Chae et al. (2026) — *TS-Cast: Deep Learning for Subsurface Ocean Reconstruction from Satellite Observations*, **Ocean Science**
- [3] Li et al. (2022) — *Subsurface Temperature Reconstruction for the Global Ocean Using Satellite Observations and Deep Learning*, **Remote Sensing**
- [4] Peng et al. (2025) — Physics-guided subsurface reconstruction using feature fusion and vertical/EOF information

Data: Copernicus GLORYS12V1 • Copernicus Marine satellite products • INCOIS Argo

Satellite surface inputs → subsurface reconstruction + predicted error
Chae et al., 2026 — TS-Cast

